

How Many People Live in Allegheny County?

One question, a shelf of published answers, and how to say which one you mean

84-355 Democracy's Data

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Ask the Census Bureau how many people live in Allegheny County and you will get an answer. Ask again somewhere else and you may get a different one. Both will be official, both current, and neither wrong.

That is the subject here. The Bureau runs three population programs, not one. Each answers a slightly different version of the question. Each publishes on its own schedule, and each leaves its older answers in print. So the population of Allegheny County is not a number you look up. It is a shelf of numbers, and the skill is knowing which one to take down.

01 Three programs, three kinds of number

The decennial census counts. Every household in the country is asked, once a decade, and the answer is an actual head count on one day. For Allegheny County on 1 April 2020 that count was **1,250,578**. Nothing else the Bureau publishes is a count. How the counting works, and who it misses, is in the decennial chapter.

The American Community Survey asks a sample. About 3.5 million addresses a year, continuously. It publishes an estimate rather than a count, and it pools either one year or five years of answers together before publishing. The survey chapter covers why there are two windows.

The Population Estimates Program calculates. Nobody is asked anything. The last census count is carried forward with births, deaths and moves, year by year, to produce a figure for 1 July. The newest one for Allegheny County is **1,225,035**. The arithmetic is in the estimates chapter.

A count, a survey and a calculation will not agree, and there is no reason they should. They are answering different questions about a county whose population was changing the whole time.

02 One file, four tables, two populations

The ambiguity starts before you leave a single file. The 2020 census publishes its county numbers in a redistricting file with four tables in it, named P1 through P4. Reading the labels, you might expect four populations. There are two.

Tables **P1** and **P2** both total everybody: 1,250,578 people. They differ only in how they break that total apart, P1 by race and P2 by Hispanic origin. Tables **P3** and **P4** do the same job for the population aged 18 and over, which is **1,020,183** — 81.6% of the whole.

So “the 2020 census population of Allegheny County” is already two different numbers, 1,250,578 and 1,020,183, and they are 230,395 people apart. Which one you want depends on the question. Apportioning seats in Congress uses everybody. Drawing districts of equal voting strength often uses the 18-and-over figure instead. Nothing in the file’s name tells you that.

Before you read on, commit to two numbers. The Bureau currently has several separate figures in print for the population of this one county. **How many do you think there are, and how far apart are the highest and the lowest?** Write both down.

03 Every answer currently in print

Here is the shelf. Every row is a figure the Census Bureau publishes today for one county, and every one of them is current in the sense that nothing has been withdrawn.

Program	Product	Counts	Refers to	Population
Population Estimates	vintage 2025 base	everybody	1 April 2020	1,250,581
Decennial census	2020 P.L. 94-171, table P1	everybody	1 April 2020	1,250,578
American Community Survey	five-year 2017-2021, table B01003	everybody	2017-2021 average	1,246,116
American Community Survey	five-year 2018-2022, table B01003	everybody	2018-2022 average	1,245,310
American Community Survey	five-year 2019-2023, table B01003	everybody	2019-2023 average	1,240,476
American Community Survey	five-year 2020-2024, table B01003	everybody	2020-2024 average	1,238,177
American Community Survey	one-year 2021, table B01003	everybody	2021 average	1,238,090
Population Estimates	vintage 2021	everybody	1 July 2021	1,238,090
American Community Survey	one-year 2022, table B01003	everybody	2022 average	1,233,253
Population Estimates	vintage 2022	everybody	1 July 2022	1,233,253
American Community Survey	one-year 2024, table B01003	everybody	2024 average	1,231,814
Population Estimates	vintage 2024	everybody	1 July 2024	1,231,814

Program	Product	Counts	Refers to	Population
Population Estimates	vintage 2024	everybody	1 July 2023	1,230,138
Population Estimates	vintage 2025	everybody	1 July 2023	1,228,911
Population Estimates	vintage 2025	everybody	1 July 2025	1,225,035
American Community Survey	one-year 2023, table B01003	everybody	2023 average	1,224,825
Population Estimates	vintage 2023	everybody	1 July 2023	1,224,825
Decennial census	2010 P.L. 94-171, table P1	everybody	1 April 2010	1,223,348
Decennial census	2020 P.L. 94-171, table P3	18 and over	1 April 2020	1,020,183

19 published figures, 18 of which count everybody. Among those 18, the highest is 1,250,581 and the lowest is 1,223,348. The gap between them is **27,233 people**, or 2.18% of the county.

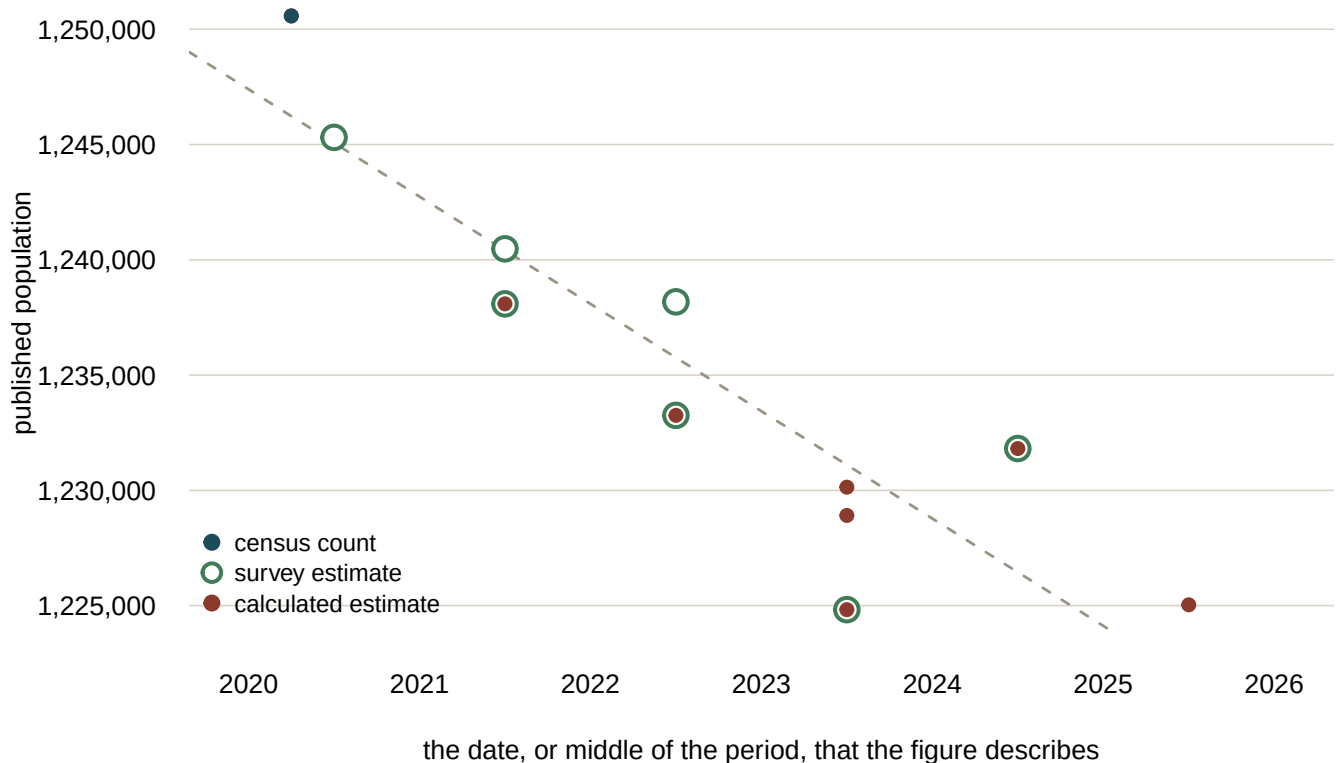


Figure 1. Every published figure that counts everybody, plotted against the date or period it describes. Survey estimates are drawn as rings and the other two programs as solid dots, so a ring with a dot inside it is two products publishing one number. At the top left the census count hides the estimates base, which sits three people away. The 2010 count is off the left edge and is left out to keep the recent figures readable.

Read the picture rather than the numbers. The figures do not scatter at random. They fall in step with the date each one describes, because the county has been losing people. The correlation between the two is -0.89 , where -1 would be a perfect straight line down.

The spread is mostly not disagreement. It is time.

That is the first practical rule. When two Census figures differ, ask what date each refers to before asking which is right.

04 The same date, published five times

The calculation has a second habit, and it catches people who have already learned the first one. **Every year the estimates program republishes its whole series, revised, back to the last census.** The label on that release is called a *vintage* – the year the Bureau did the arithmetic, not the year the number describes.

So a single date has been published over and over, with a different answer each time.

Refers to (1 July)	Times published	Lowest	Highest	Spread
2020	5	1,248,717	1,249,671	954
2021	5	1,238,090	1,245,445	7,355
2022	4	1,232,605	1,233,821	1,216
2023	3	1,224,825	1,230,138	5,313
2024	2	1,227,174	1,231,814	4,640
2025	1	1,225,035	1,225,035	0

1 July 2021 has been published 5 times, and the published value moves 7,355 people across those releases. That date is fixed. The population on it is fixed. Only the Bureau's estimate of it has changed, as later information arrived.

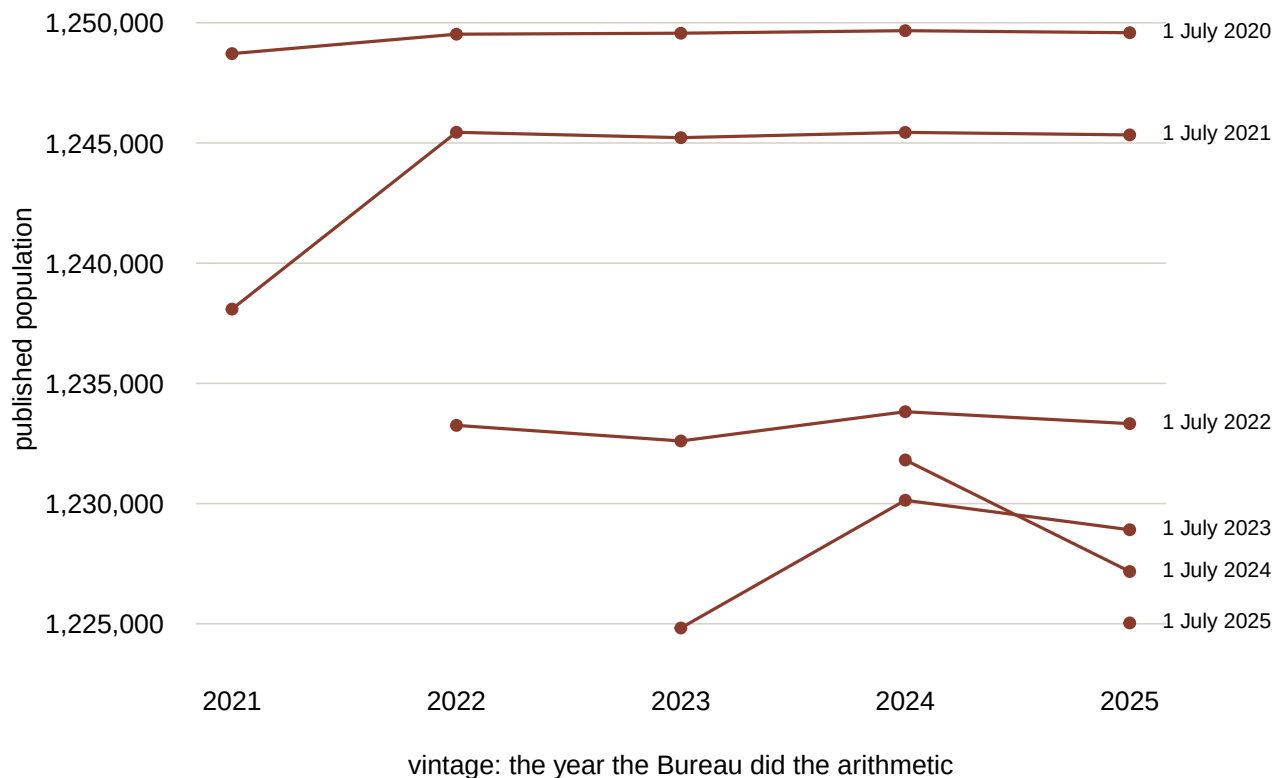


Figure 2. Each line is one reference date; each point is a separate vintage's estimate of it. A date does not settle once it is published.

Allegheny County is not unusual in this. Taking the reference date that moved most, and asking the same question of every county in the country: **0 of 3,135 counties kept the**

same published value across every vintage. The middle county moved by 0.27%, and 255 counties moved by more than one per cent.

The practical rule follows. **Cite the vintage, not just the year.** “The 2021 population estimate” names five different numbers.

05 Two of these rows are the same number

Look back at the catalog and something odd is sitting in it. The one-year survey estimate for 2024 and the vintage 2024 calculation are both 1,231,814. That is not a coincidence, and it is not rounding.

Survey window	Compared with	Counties	Identical
one-year 2021	estimates vintage 2021, 1 July 2021	830	829
one-year 2022	estimates vintage 2022, 1 July 2022	837	836
one-year 2023	estimates vintage 2023, 1 July 2023	843	842
one-year 2024	estimates vintage 2024, 1 July 2024	850	849
five-year 2020-2024	estimates vintage 2025, 1 July 2021	3,144	7
five-year 2020-2024	estimates vintage 2025, 1 July 2022	3,144	17
five-year 2020-2024	estimates vintage 2025, 1 July 2023	3,144	9
five-year 2020-2024	estimates vintage 2025, 1 July 2024	3,144	4
five-year 2020-2024	estimates vintage 2025, 1 July 2025	3,144	5

The one-year survey figure for a county’s total population is the calculated estimate for that year. Not close to it. The same number, in 3,356 of 3,360 comparisons across four years of files. The Bureau forces its survey totals to agree with the estimates before publishing them, so asking the survey for a county’s population returns the calculation wearing a survey’s label.

One county in the whole country is an exception. The two figures for Maui County, Hawaii differ in every year tested, by at most 61 people. Why that county and no other is not something the files say.

The five-year figure behaves differently, and the contrast is what makes the result above worth trusting. It matches no published estimate for any year: the best any single year manages is 17 counties out of 3,144. Five years of answers get pooled, so the total is tied to five years of estimates at once and lands on none of them.

This matters when you are tempted to average. Two sources that agree feel like confirmation. Here, one of them is the other, and averaging them counts a single number twice.

06 Allegheny is not the case, it is the illustration

The same shelf exists for every county. Taking the current figures from all three programs, county by county:

- **3,222 counties** carry at least one figure from each.
- **0 of them** have every figure agreeing.
- The middle county's figures span **3.47%** of its largest.
- **1,081 counties** span more than five per cent.

Read those numbers with the first rule in mind. Much of that spread is real population change between the dates the figures refer to, not disagreement about a shared moment. That is the point rather than a caveat. A figure with no date attached is not a population, and the country's counties will not let you pretend otherwise.

07 Where each one lives

Three addresses serve all of this, and they do not carry the same things.

Address	Hands you	You need	Carries	Best for
data.census.gov	one table at a time, drawn in a browser	nothing	a search over the published tables; it cannot be scripted	looking one number up, once, by hand
api.census.gov	the cells you name, as JSON	a free key, for data, since 2026	1,798 datasets; county estimates stop at vintage 2021	a few named numbers inside a program
www2.census.gov	the published file, whole	nothing	the files themselves, county estimates through vintage 2025	a whole product, or a vintage the other two have dropped

The interface publishes an index of its own holdings, and that index is public and needs no key. It is the honest way to ask a route what it carries, and it also shows what a route is missing.

Product	Path	Vintages	Earliest	Latest
Decennial redistricting file	dec/pl	3	2000	2020
Survey, five-year	acs/acs5	16	2009	2024
Survey, one-year	acs/acs1	19	2005	2024
Population estimates	pep/population	6	2015	2021

County population estimates stop at vintage 2021 there. The 4 newer vintages exist, and they are the ones in the table above, but they are published as files and not through the interface. A script that reaches for the newest estimates by that route will quietly get an old one.

Two more things are worth knowing before you go looking.

There is no 2020 one-year survey. Response rates during the pandemic were too low for the Bureau to publish the standard series, so it did not. The address for that file answers 404, and the year is missing from the index as well. A gap in a series is not always an error in your code.

The one-year survey does not cover most counties. It publishes 861 county rows against the five-year file's 3,222, so **2,361 counties appear in one and not the other**. A sample thin enough to cover a small county reliably does not exist in a single year.

08 Naming a number so somebody else can find it

Everything above collapses into one habit. **Name the product, the vintage and the date before you name the number.**

"Allegheny County has about 1.23 million people" is not checkable. 7 rows of the catalog round to that figure. They refer to 6 different periods and hold 5 distinct values, and 2 of those values each appear twice. A reader cannot get back to what you used.

These are checkable:

- the 2020 census counted **1,250,578** people there, on 1 April 2020
- the 2020–2024 five-year survey estimates **1,238,177**
- the vintage 2025 estimate for 1 July 2025 is **1,225,035**

Each one names a file somebody else can open. That is the whole difference.

The same discipline applies to boundaries, which have vintages too. A county whose lines moved between two releases will fail to join, silently, and the geography chapter works through what that does to a map.

09 What this chapter cannot tell you

It does not say which figure is right. The question has no answer in that form. A count on one day, a five-year average and a calculation for last July are three different quantities. Which you want follows from what you are asking, and nothing in the data can choose for you.

It does not explain how the calculation is made. That the published value for a fixed date moves across vintages is shown here. Why it moves is not. Which records arrived late, and which assumption was revised, is the subject of the estimates chapter.

It follows one table. Total population is a single table out of thousands the Bureau publishes. Income, age, language and the rest each have their own products and their own vintages, and the same discipline applies to all of them.

The shelf will have changed by the time you read this. A new vintage arrives each spring and a new survey release each winter. The figures here were fetched in August 2026. The pattern is durable; the particular numbers are not, and a rebuild that returns different values has not gone wrong.

10 Sources

- U.S. Census Bureau. *2020 Census P.L. 94-171 Redistricting Data Summary File*, Pennsylvania, tables P1 through P4. https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/
- U.S. Census Bureau. *American Community Survey*, table B01003, one-year and five-year releases for 2021 through 2024, from the table-based summary file. https://www2.census.gov/programs-surveys/acs/summary_file/
- U.S. Census Bureau. *County Population Totals*, vintages 2021 through 2025, and the 2010-2020 series that carries the 2010 count. <https://www2.census.gov/programs-surveys/popest/datasets/>
- U.S. Census Bureau. *Census Data API discovery index*, the public catalog of every dataset the interface serves. <https://api.census.gov/data.json>
- The control relationship that makes a one-year survey total equal to a population estimate is documented by the Bureau in *Understanding and Using American Community Survey Data*, and taken apart from the survey's side in the ACS chapter.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`api_vintages.csv`, `catalog.csv`, `pep_revisions.csv`, `pep_same_date.csv`,
`route_overlap.csv`, `routes.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `catalog.csv` lists every published answer to a single question. Sort by population. How far apart are the highest and the lowest, and are they even measuring the same thing?
- `pep_same_date.csv` shows one date published several times with a different number each time. Look at `spread`. Which of them is the population of that day?
- `pep_revisions.csv` follows one year's estimate as later vintages revise it. Do the revisions drift steadily one way, or wobble around?
- `routes.csv` sets out three ways to reach the same number and what each one costs you. Decide the question you would actually ask, then say which route you would take.

REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original sources, without any starting files. Please do the following and show your work.

THE SOURCE. The U.S. Census Bureau, which publishes more than one answer to the question "how many people live in this county?" The county followed throughout is Allegheny County, Pennsylvania, FIPS 42003. Everything below is free, needs no key, and lives on www2.census.gov.

THE COUNT. The 2020 redistricting file for Pennsylvania:

https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File-PL_94-171/Pennsylvania/pa2020.pl.zip

Pipe-delimited, no headers. Counties are summary level 050 in the geographic header file, which joins the data segments on LOGRECNO. Segment 1 holds tables P1 and P2; segment 2 holds P3 and P4. In each segment the first table's first cell is field 6 and the second table's first cell is field 77.

THE SURVEY. Table B01003 is total population. Fetch both windows for 2021 through 2024, eight files in all, from

https://www2.census.gov/programs-surveys/acs/summary_file/YYYY/table-based-SF/data/

under 1YRData/acsdt1yYYYY-b01003.dat and

5YRData/acsdt5yYYYY-b01003.dat. Pipe-delimited with a header; a GEO_ID beginning 0500000US is a county. Try the 2020 one-year file too, and record what its address returns.

THE CALCULATION. County population totals, one file per vintage, from <https://www2.census.gov/programs-surveys/popest/datasets/> - 2020-YYYY/counties/totals/co-estYYYY-alldata.csv for YYYY 2021 through 2025

- 2010-2020/counties/totals/co-est2020.csv, which carries the 2010 census count

Each vintage republishes every year back to 2020, so vintage 2025 contains POPESTIMATE2020 through POPESTIMATE2025.

THE INDEX. <https://api.census.gov/data.json> is the programming interface describing its own holdings, keyless. Each record has a c_dataset path and a c_vintage year.

WHAT TO BUILD.

1. A catalog: one row per published figure for Allegheny County, with the program, the product, what it counts, and the date or period it refers to. Keep the 18-and-over figure separate from the figures that count everybody.
2. For one fixed reference date, every vintage's estimate of it, so the revisions are visible side by side.
3. Two comparisons across all counties: the one-year survey total

against the estimate for the same year, and the five-year survey total against every estimate year.

4. From the index, how many vintages of each product it carries and the newest vintage of county estimates it offers.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- Allegheny County counted 1,250,578 people in 2020, of whom 1,020,183 were 18 or over
- 19 published figures in the catalog, 18 of them counting everybody, with 14 distinct values and a spread of 27,233 people
- 1 July 2021 has been published 5 times, the values 7355 apart, and 0 of 3,135 counties kept one value across every vintage
- the one-year survey total equals the estimate in 3,356 of 3,360 county comparisons, the exception being Maui County, Hawaii
- the five-year total matches at best 17 of 3,144 counties for any single estimate year
- the index lists 1,798 datasets and stops at vintage 2021 for county population estimates
- the 2020 one-year file's address returns 404

A new vintage lands each spring and a new survey release each winter, so a rebuild done later will find more rows and different values. The pattern is the finding; the particular numbers are dated.